



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING) Academic Year: 2025-26

Name of Student: Prajwal Dhanawade

Semester: VI

Class / Div / Branch: T.E./ B1/ CSE(AI&ML)

Subject: SEPM

Name of Instructor: Shruthi Nair

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Roll No: 75

Date of Performance:

Date of Submission: 19/01/2026

Experiment No. 1

Aim: To study DevOps concepts including its principles, practices, and the roles and responsibilities of a DevOps engineer.

Lab Outcomes: To understand the fundamentals of DevOps engineering and be fully proficient with DevOps terminologies, concepts, benefits, and deployment options to meet your business requirements

Team Details

- **Team Number:** _____
- **Student Names & Roll Numbers:**
 1. Pranay Bhare - 71
 2. Sanidhya Pandey - 83
 3. Om Brahmavale - 73
 4. Prajwal Dhanawade - 75

Assigned Roles (One Role per Student)

Student Name	Role Assigned
Pranay Bhare	Developer
Sanidhya Pandey	DevOps Engineer
Om Brahmavale	Operations Engineer
Prajwal Dhanawade	Observer / Manager

Activity 1: Traditional vs DevOps Workflow

Scenario

You are asked to **release a new feature for a college website**.

Task A: Traditional Approach (No DevOps)

List the steps followed and problems faced.

- Steps followed:
 1. The manager tasks the developer to add a news panel feature for the college website.
 2. Developer will write the code for the proposed feature.
 3. Code is then handed down to the operations engineer.
 4. Operations team then manually deploys the code **without** connecting with the developer.
 5. Testing is done after deployment
- Problems observed:
 1. Poor coordination between teams
 2. Manual deployment causes errors.
 3. Slow testing and release process
 4. Late detection of bugs
 5. Difficult rollback in case of failure.

Task B: DevOps Approach

List how the same task is handled using DevOps.

- Steps followed:
 1. Manager, developer, devops engineer and operations engineer plan together.
 2. Developer writes code and pushes to a shared repository
 3. DevOps engineer sets up CI/CD pipeline.
 4. Automated build and testing are performed
 5. Continuous deployment is done automatically
 6. Operations engineer monitors the system performance.
- Improvements achieved:
 1. Faster and frequent releases.
 2. Better teamwork and deployment.
 3. Automated testing and deployment.
 4. Reduced failures and downtime
 5. Faster issue detection and recovery.

Activity 2: Role & Responsibility Mapping

Task	Developer	DevOps Engineer	Operations
Write application code	✓	x	x
Coordinate between teams	x	✓	✓
Ensure smooth deployment	x	✓	✓
Monitor system performance	x	✓	✓
Handle failures/issues	x	✓	✓

Activity 3: DevOps Lifecycle Mapping

Write the DevOps lifecycle stages in order and mention the responsible role.

Stage	Responsibility
Plan	Manager
Code	Developer
Build	DevOps Engineer
Test	DevOps Engineer
Release	DevOps Engineer
Monitor	Operations Engineer

Activity 4: Failure Scenario Analysis

Scenario

During online exam result declaration, the website crashes.

Answer the following:

1. What could be the reason?
ANS: High Traffic, server overload
2. Which team is responsible?
ANS: DevOps and Operations team
3. How can DevOps prevent this issue?
ANS: Load testing before release & Auto-scaling servers

Activity 5: Team Retrospective

Discuss within your team and write short answers.

1. What did your team do well?
ANS: Completed tasks on time.
2. What problems did you face?
ANS: Lack of DevOps knowledge and understanding of new tools
3. How can teamwork be improved?
ANS: Better communications.

Learning Outcomes

After completing this activity, I understand:

- What DevOps means: DevOps is a way of working where multiple teams collaborate and work together.
- Role of a DevOps Engineer: Automaton, testing, monitoring
- Importance of teamwork in DevOps: Teamwork helps in faster delivery and fewer errors.

Conclusion:

The experiment helped in understanding the DevOps concepts including its principles, practices, and the roles and responsibilities of a DevOps engineer